

```
import pandas as pd

df = pd.read_csv("Dataset .csv")
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswana Pula(
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswana Pula(

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Restaurant ID                        9551 non-null   int64
1   Restaurant Name                      9551 non-null   object
2   Country Code                        9551 non-null   int64
3   City                                9551 non-null   object
4   Address                             9551 non-null   object
5   Locality                            9551 non-null   object
6   Locality Verbose                    9551 non-null   object
7   Longitude                           9551 non-null   float64
8   Latitude                           9551 non-null   float64
9   Cuisines                            9542 non-null   object
10  Average Cost for two                 9551 non-null   int64
11  Currency                            9551 non-null   object
12  Has Table booking                    9551 non-null   object
13  Has Online delivery                 9551 non-null   object
14  Is delivering now                    9551 non-null   object
15  Switch to order menu                 9551 non-null   object
16  Price range                         9551 non-null   int64
17  Aggregate rating                     9551 non-null   float64
18  Rating color                        9551 non-null   object
19  Rating text                         9551 non-null   object
20  Votes                              9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
```

memory usage: 1.5+ MB

df.isnull().sum()

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

```
df = df.drop(columns=[
    "Restaurant ID",
    "Restaurant Name",
    "Address",
    "Locality",
    "Locality Verbose",
    "Longitude",
    "Latitude",
    "Rating color",
    "Rating text",
    "Switch to order menu"
])
```

```
df["Cuisines"] = df["Cuisines"].fillna("Unknown")
```

```
yes_no_cols = [
    "Has Table booking",
    "Has Online delivery",
    "Is delivering now"
]

for col in yes_no_cols:
    df[col] = df[col].map({"Yes": 1, "No": 0})
```

```
df = pd.get_dummies(df, columns=["City", "Cuisines", "Currency"], drop_first=True)
```

```
X = df.drop("Aggregate rating", axis=1)
y = df["Aggregate rating"]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.tree import DecisionTreeRegressor

model = DecisionTreeRegressor(max_depth=10, random_state=42)
model.fit(X_train, y_train)
```

DecisionTreeRegressor ⓘ ?

```
DecisionTreeRegressor(max_depth=10, random_state=42)
```

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DecisionTreeRegressor ⓘ ?

```
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```

```
pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

mse = mean_squared_error(y_test, pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, pred)

print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
MSE: 0.11463299078441638
RMSE: 0.3385749411643106
R2 Score: 0.9496364228284034
```

```
importance = pd.Series(model.feature_importances_, index=X.columns)
importance.sort_values(ascending=False).head(10)
```

	0
Votes	0.977648
Country Code	0.007614
Average Cost for two	0.002326
Currency_Brazilian Real(R\$)	0.001455
Has Online delivery	0.000903
Currency_Dollar(\$)	0.000711
Cuisines_Pizza, Fast Food	0.000613
City_New Delhi	0.000611
Has Table booking	0.000489
City_Noida	0.000419

dtype: float64

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